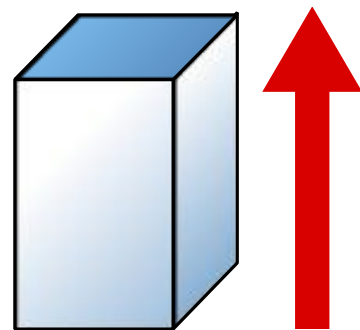


DESIGN GUIDELINES

AUTODESK INVENTOR 2012

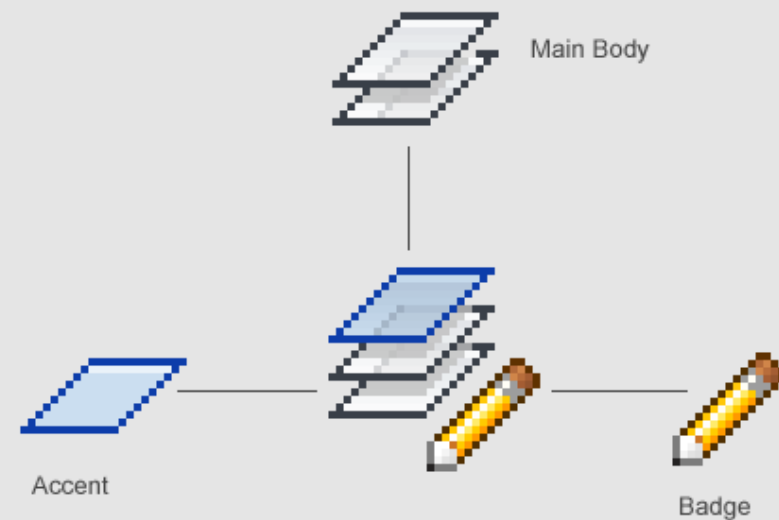
ICONS



Elements of an Icon

Three Elements of an Icon

The foundation of our icons is the main body, accents and badges enhance their meaning in a systematic way.



1. Main Body

The main body is the basic shape of an icon. it can be thought of as the background on which more vibrant elements sit. some icons only have main body such as the Gear icon.



2. Accent

Accents are spot color or an overlay that is applied to main body to enhance meaning and communicate change. Accents include pre-and post states, selection, badging and grouping. A small set of colors are reserved to communicate specific types of operations.



3. Badges

Badges are separate elements laid on top of elements below. In addition to verb-noun pairs, badges enable icon designers to create implied verb-noun pairs such as create new layer.



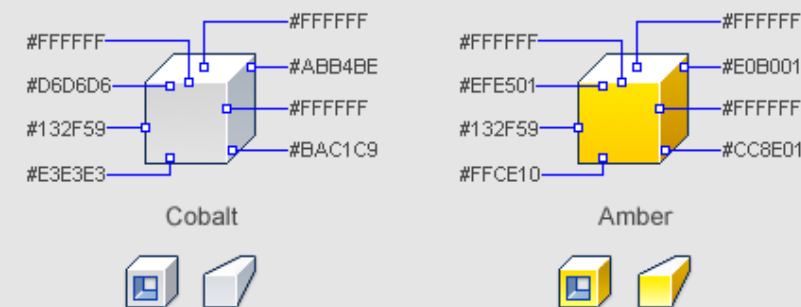
Color

Use the right color palette

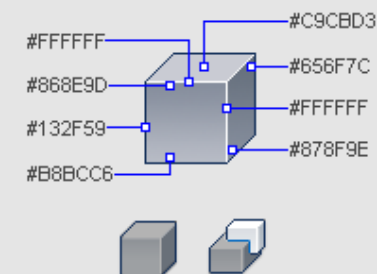
There are many types of the element; each element should use the corresponding color palette. The main elements include main body, accent and badge; and there are lines (1D), flat (2D) and three dimensional (3D) types for each element. There are two color themes for Main Body: Cobalt & Amber for the icons. Use the standard or variation color scheme except the real world items.

1. Main Body

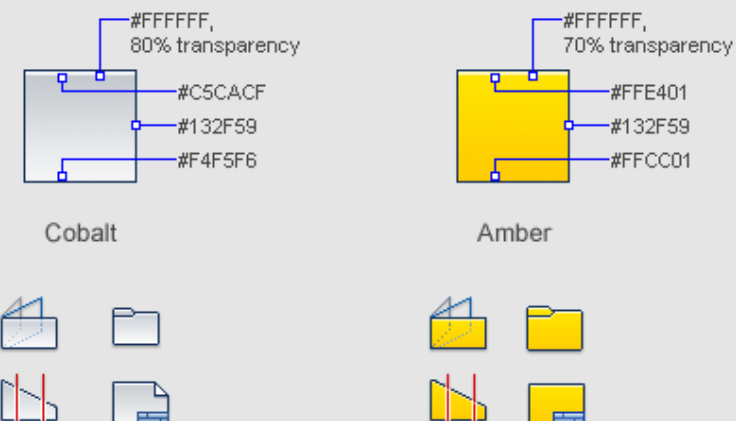
1) Standard Color \ Three Dimensional (3D)



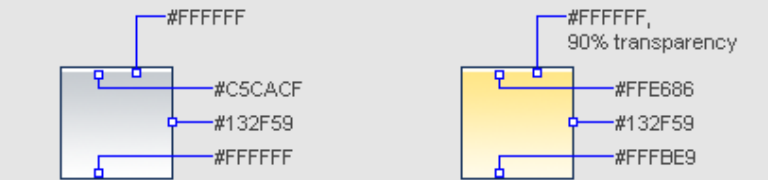
2) Variation Color \ Three Dimensional (3D)



3) Standard Color \ Flat (2D)



Color



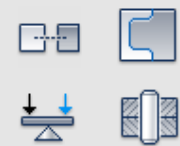
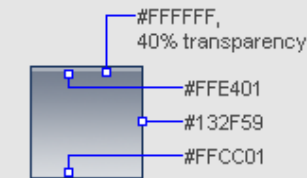
Cobalt
Small body



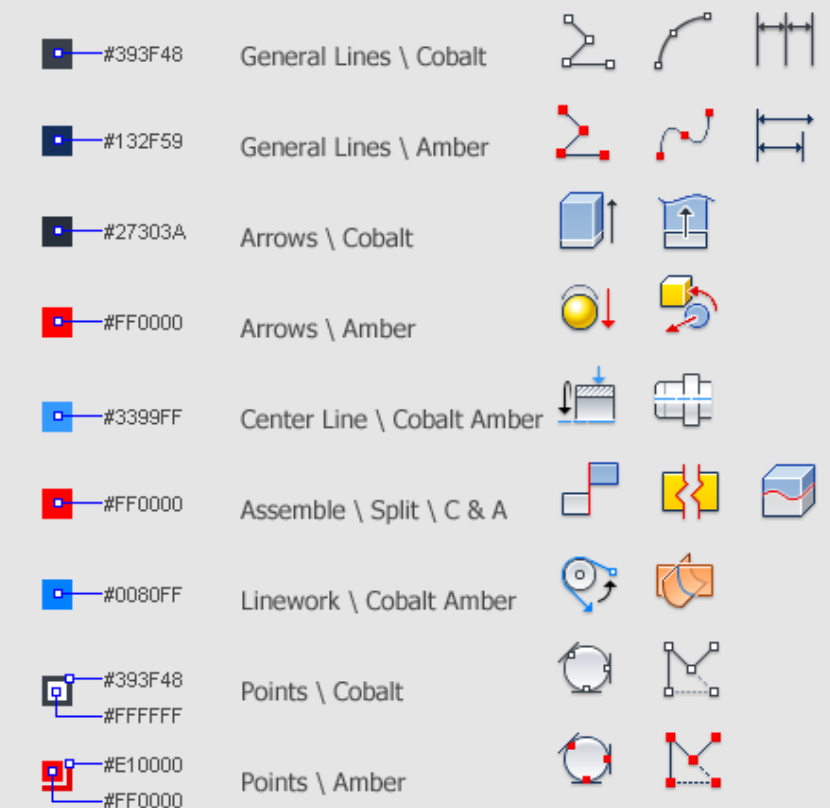
Amber
Balloons & Tables & Datum



4) Variation Color \ Flat (2D)



5) Standard Color \ Lines & Points (1D)

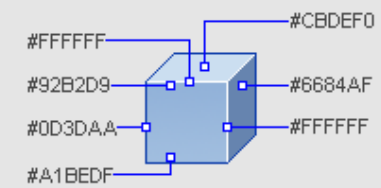




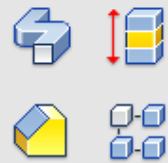
Color

2. Accents

1) Pre & Poste & Modified

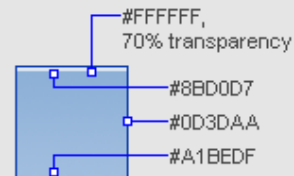


Three Dimensional (3D)



#3399FF Lines (1D)

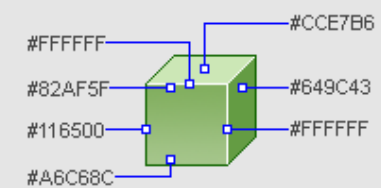
#3399FF Points



Flat (2D)



2) Selections

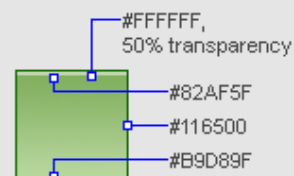


Three Dimensional (3D)

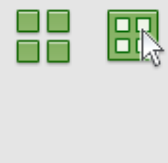


#116500 Lines (1D)

#116500 Points



Flat (2D)

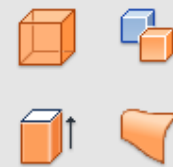


Color

3) Surfaces



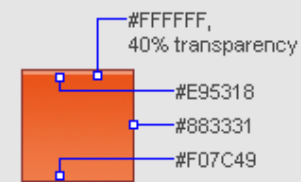
Three Dimensional (3D)



Flat (2D)



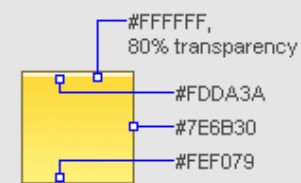
4) Welded & Interferences & Intersections



Flat (2D)



5) Soldered Connections



Flat (2D)



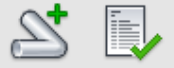
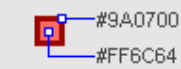
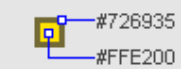
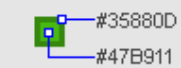
Color

3. Badges

Any color can be used for a badge. Since they are typically small, they require a high amount of saturation. For badges that are 7x7 px and smaller, use 100% saturation. For badges larger than 7x7 px, use between 80% and 90% saturation. To maintain consistency, you must use badges that have been approved by the Visual Design Group. If a badge does not exist in our database, you may create the badge using the design rules and send it to the VDG for approval and we will then enter it in our database.

1) Indicating Status

When applying color to status symbols, follow the saturation rule above.



4. Real World Items

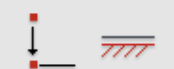
1) Additional colors may be used in real world items. Not more than two additional colors within an icon may be used. The two additional colors can be picked from the entire color spectrum. Avoid unnecessary colors.

2) As the main body, they must never be more saturated than accents or badges.

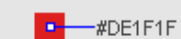


5. Other

1) Constraints & Snap-Points/Lines



2) Wires



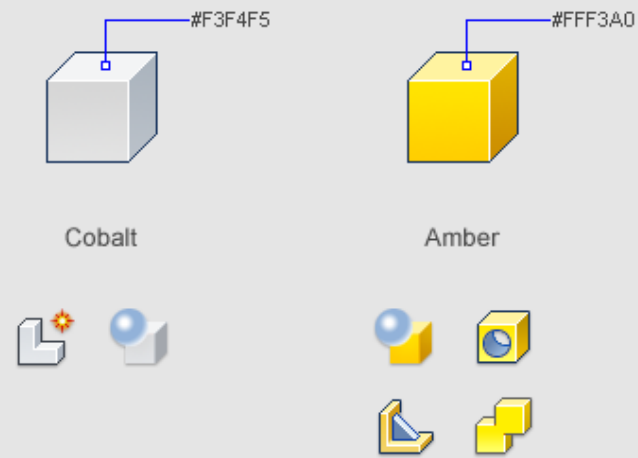


Color

6. Special

1) Highlight Edges

If need highlight the inner edges for 3D main body, or need avoid white color for top face; set the colors as follow.



2) Triad

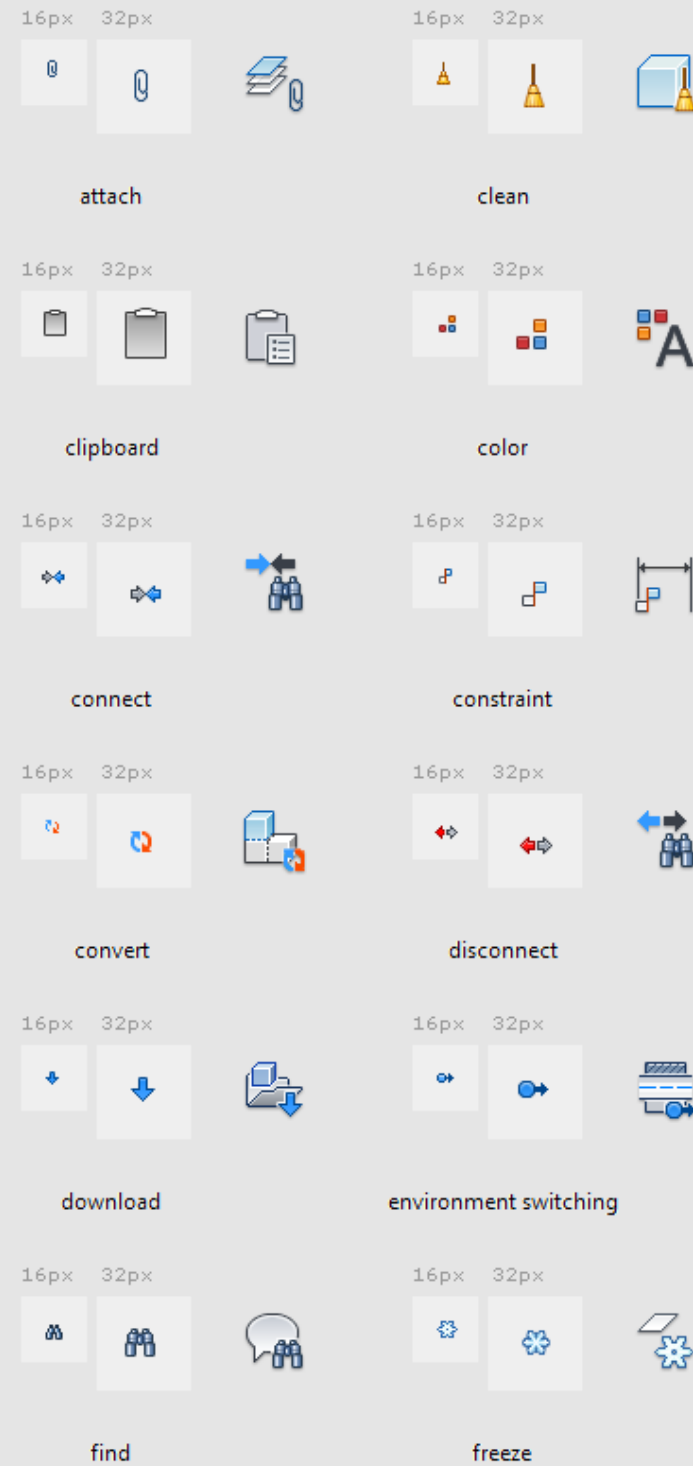
If need diffient color to show each triad, use the special color setting as follow.

- #FF0000 X axis
- #007D00 Y axis
- #0000FF Z axis

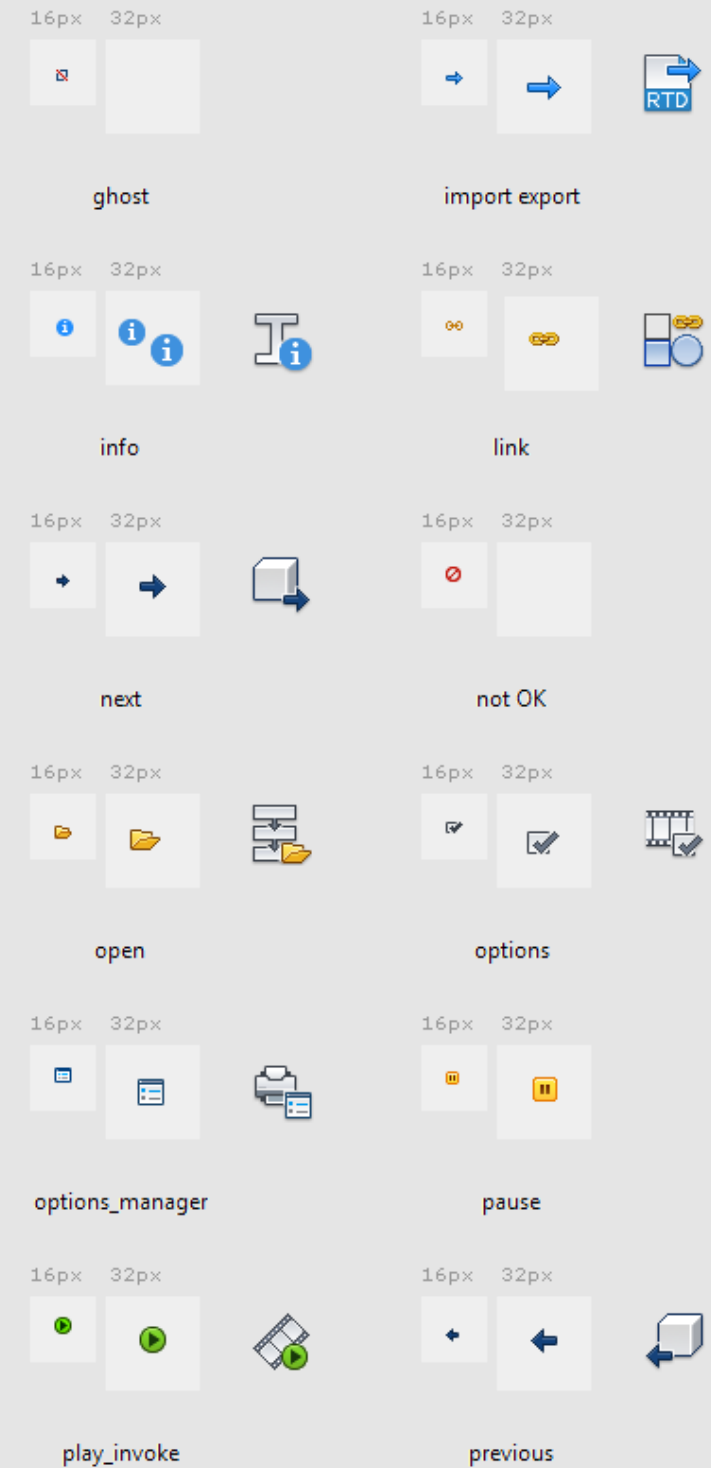


Badges

1. Approved Badges

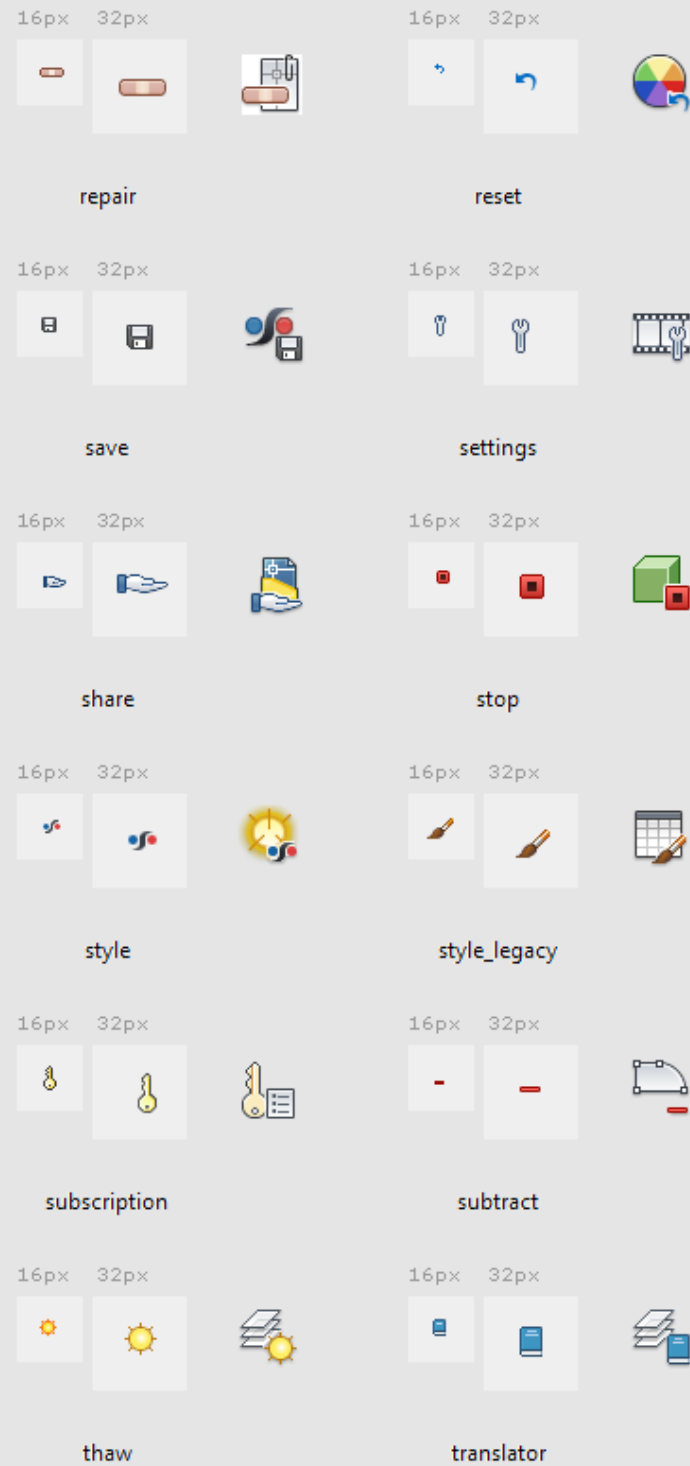


Badges

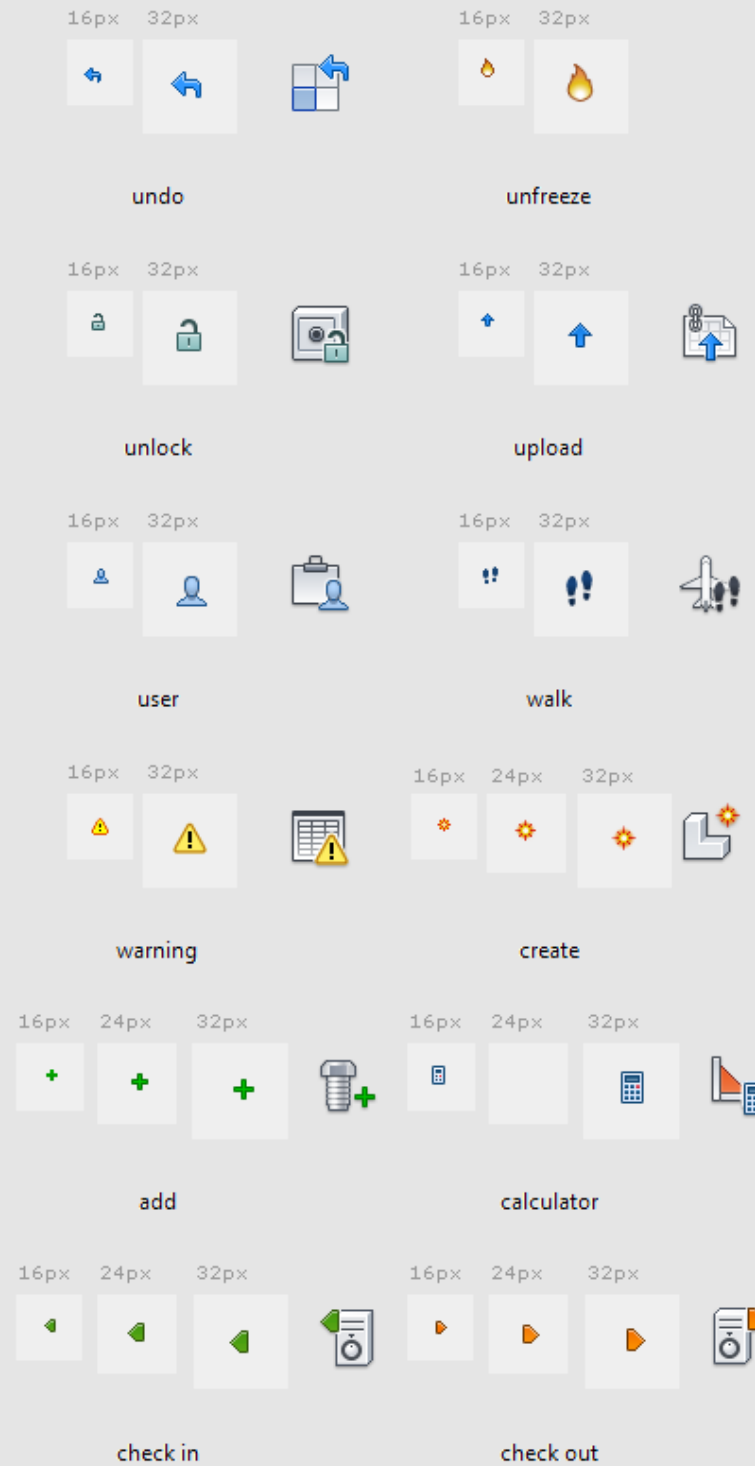




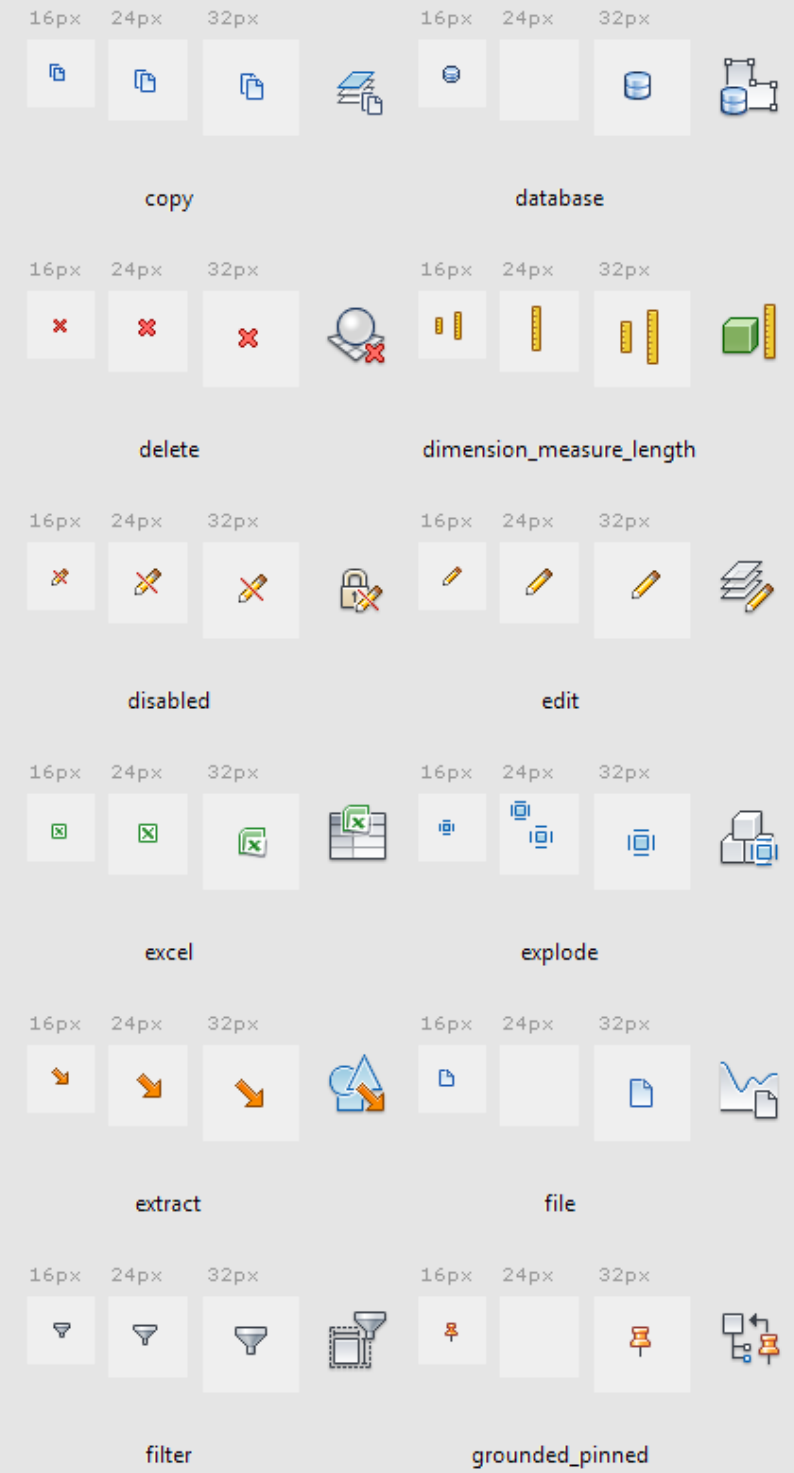
Badges



Badges



Badges



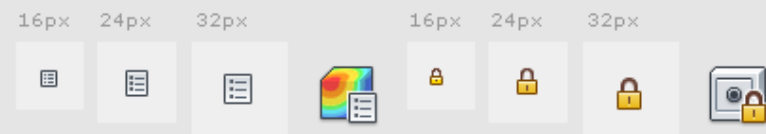


Badges



group

insert



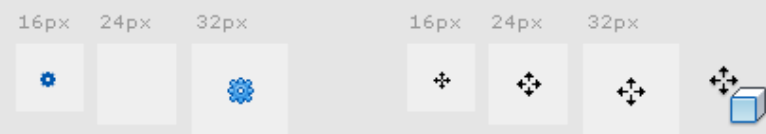
list

lock



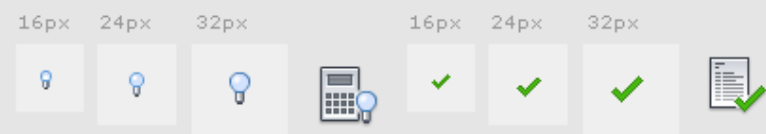
manager

match



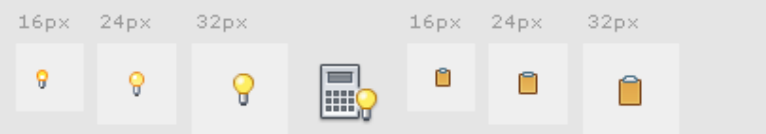
mechanical

move



off

OK_check



on

paste

Badges



pointers

properties



rotate

scale



search

shortcut



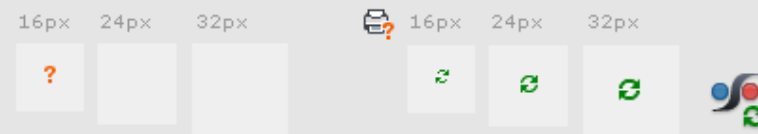
snap

stamp



standards

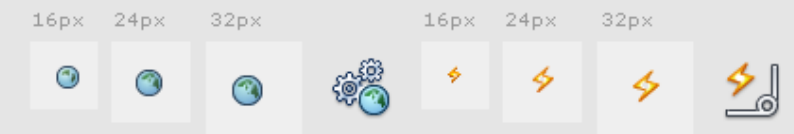
surface



unresolved

update

Badges



web.png

wizard_quick_automate.png

2. Badges Style

1) Badges are applied above all other icon elements.

2) There are 9 positions for the Badges, the below 9 icons are the example.

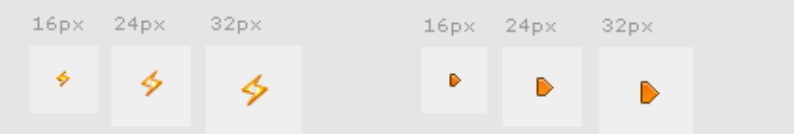


3) The Badges are set default in the lower right of an icon. You should put the badges in this position in general icon design.



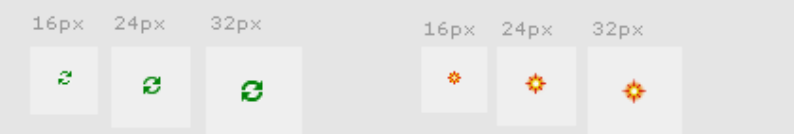
4) Some Badges are set in other default position of an icon.

(1) Upper Right



wizard_quick_automate.png

check out



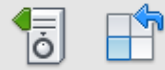
update

create



Badges

(2) Upper Left



16px 24px 32px



check in

16px 32px



undo

16px 32px

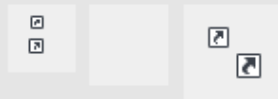


reset

(3) Lower Left



16px 24px 32px



shortcut

16px 32px



share

16px 32px

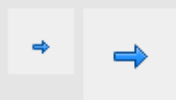


previous

(4) Left



16px 32px



import

Badges

(5) Right



16px 32px



export

(6) Upper



16px 32px



upload

(7) Lower



16px 32px



download

5) Another position should be chosen only if it helps to clarify the meaning or supplies better contrast and readability of the icon.

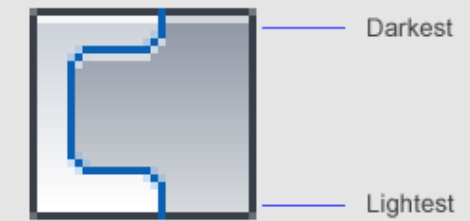


Lighting

1. Surface Fills

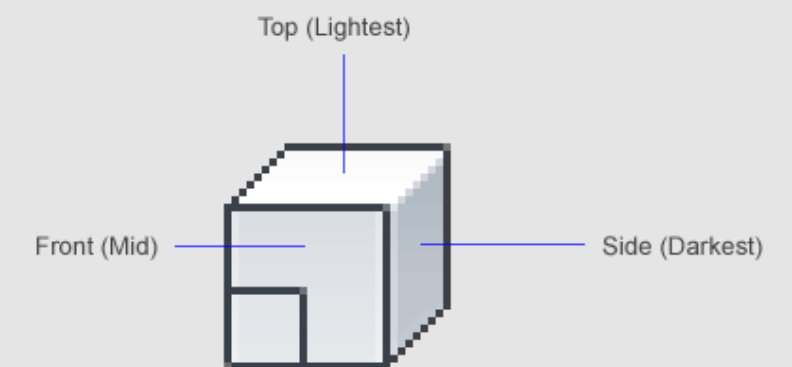
1) Flat (2D) objects

Fills are lightest at the bottom and get darker toward the top. The exact color values are ruled in Color pages.



2) Three dimensional (3D) objects

Use lightest fill for top facing areas, mid gradient for front facing areas and darkest gradient for both sides. The exact color values for oblique bimetric projection box are ruled in Color pages.



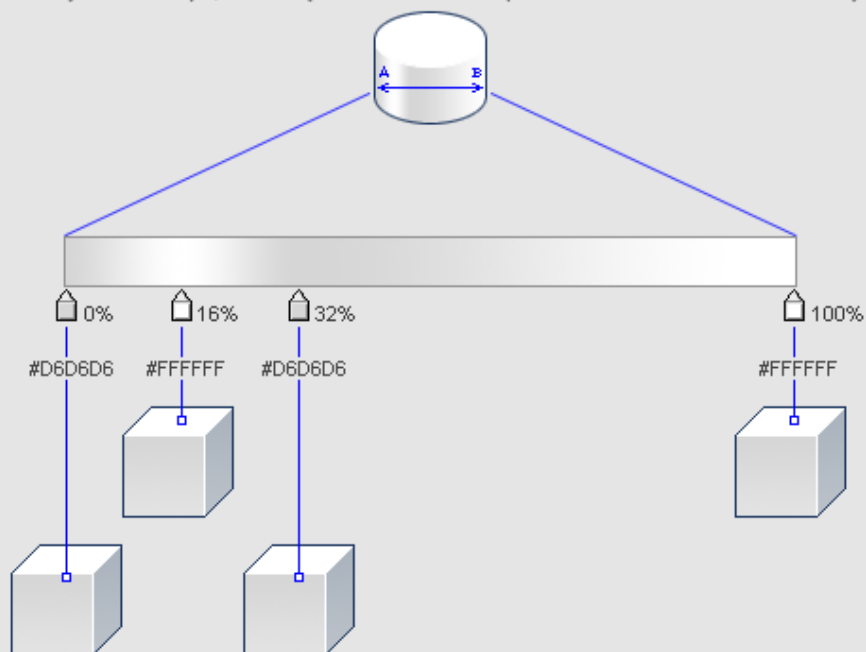
Light of Cylinder, Cone, Sphere and Torus are follow the below rules.

Lighting

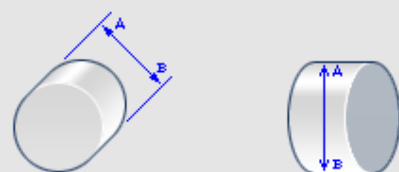
2. Cylinders

The icon light source originates from the top-left of the cylinder. So the cylinder faces are gradient filling effect. There are three place orientation cylinders and the cylinder faces are the same gradient values, only the direction of the gradient are different.

1) Main body \ Cobalt (all the colors are picked form standard color box)



The direction of gradient for other two orientation cylinders.



2) Main body \ Amber



3) Variation



Lighting

4) Pre & Post & Modified



5) Selections



6) Surfaces



8) Highlight Edge (Pastel) \ Cobalt (Use to stand for Main body cobalt)



9) Highlight Edge (Pastel) \ Amber (Use to stand for Main body amber)

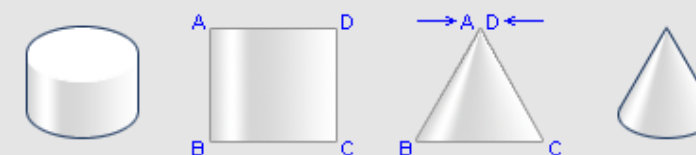


3. Cone

Cone is base Cylinder face's gradient filling, but add a perspective effect.

Lighting

1) Main body \ Cobalt



2) Main body \ Amber



3) Variation



4) Pre & Post & Modified



5) Selections



6) Surfaces



7) Pastel \ Cobalt



8) Pastel \ Amber



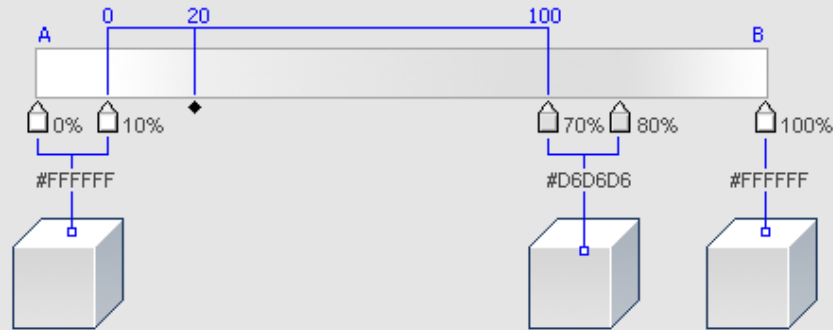
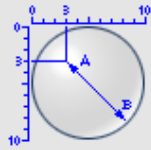
4. Sphere

The icon light source originates from the top-left of the icon. So use a 45 degree Radial Gradient from top-left to bottom-right.



Lighting

1) Main Body \ Cobalt



2) Main Body \ Amber



3) Variation



4) Pre & Post & Modified



5) Selections



6) Surfaces

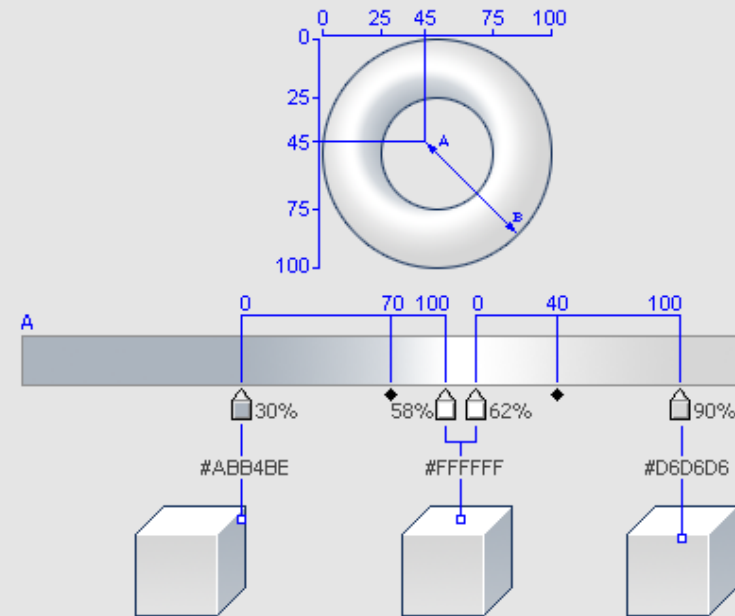


Lighting

5. Torus

The icon light source originates from the top-left of the icon. So also use a 45 degree radial gradient from top-left to bottom-right. Some kinds like the sphere, but different positions.

Example:



1) Main Body \ Cobalt



2) Main Body \ Amber



3) Variation



4) Pre & Post & Modified



5) Selections



6) Surface



Lighting



6. Real World Items

The direction of the gradients can be reversed to achieve more realistic Rendering.



7. Highlights

Add a white 1px highlight at the darker end of the surface gradient. For the specified colors, the highlights specified in Color section too.



1) Highlight Circle

There are two type Circle highlights: Crescent highlight and Linear highlight. You can Chose each one to use by referring icon practical conditions.

(1) Linear

Create Steps:

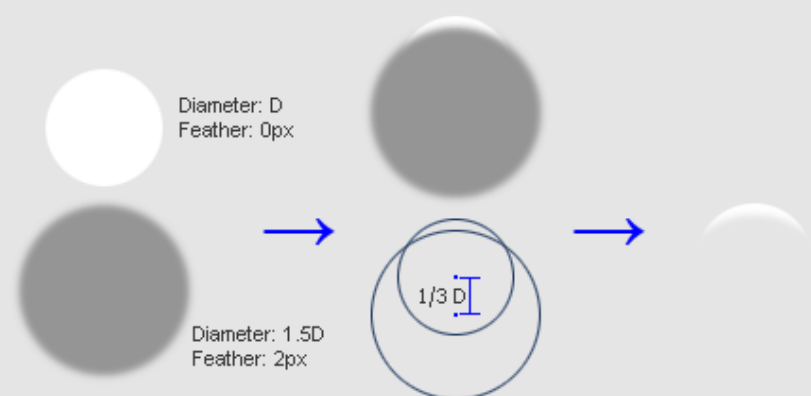
1. create a circle and the arc (1px width).
2. copy the arc and change it to white color as the highlight arc.
3. move down the highlight arc 1px then delete the lower half.
4. change the highlight arc 70~100% transparency.
5. set the highlight arc layer between circle layer and arc layer.





Lighting

(2) Crescent

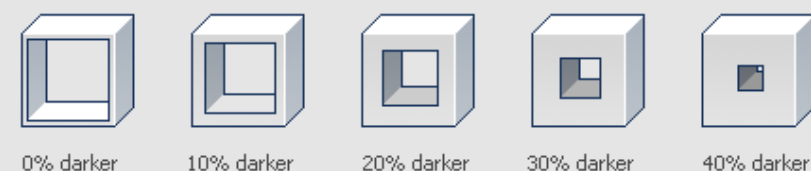
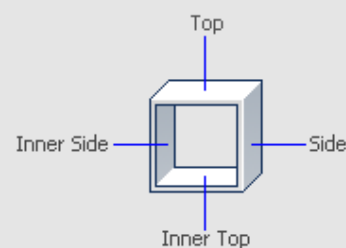


8. Void Three dimensional (3D) object

Void 3D objects are the "negative" object. They are the shadows of the models.

1) Box

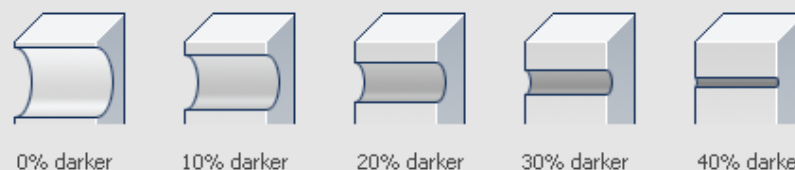
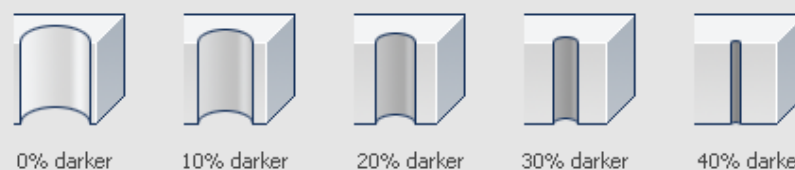
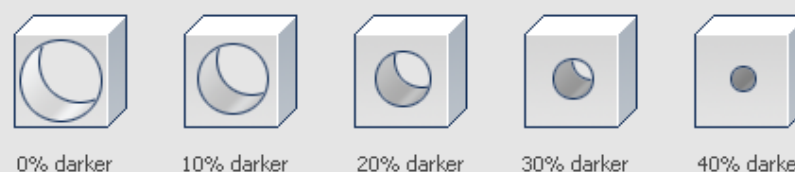
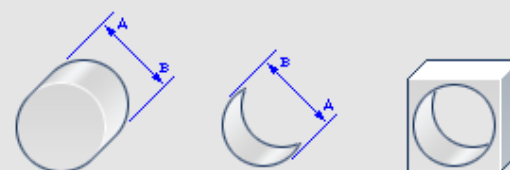
The inner top face and inner side face is the same gradient values as top faces and side faces of the box if the void size is large; when the void size is get smaller, the gradient should more darker. There are not precision dark values, just make them in consistency.



Lighting

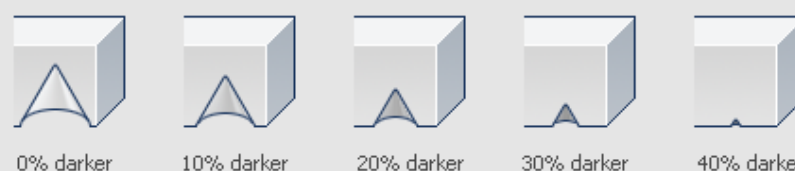
2) Cylinder

The hole (inner cylinder) faces are the same gradient values as cylinder faces but the gradient directions are reverse.



3) Cone

The same as cylinders, fill the reverse cone gradient on the face.



Lighting

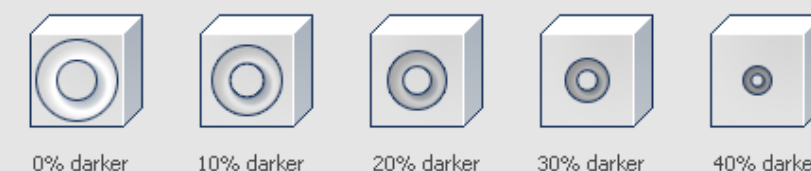
4) Sphere

The same as cylinder, fill the reverse sphere gradient on the face.



5. Torus

The same as cylinder, fill the reverse torus gradient on the face.





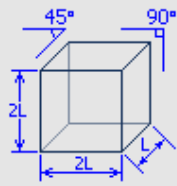
Perspective

There are few kinds of perspectives use in Inventor icons. They are use to represent the 3D object usually.

1. Axonometric

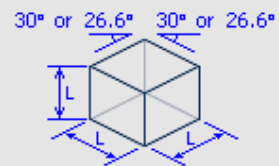
1) Oblique bimetric projection

The oblique bimetric projection is apply for 3D box and also is the general projection type in Inventor.

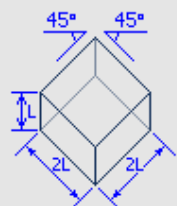


2) Isometric projection

The isometric projection usually apply for cylinder and cone, also some special box can use isometric projection if it preferably.



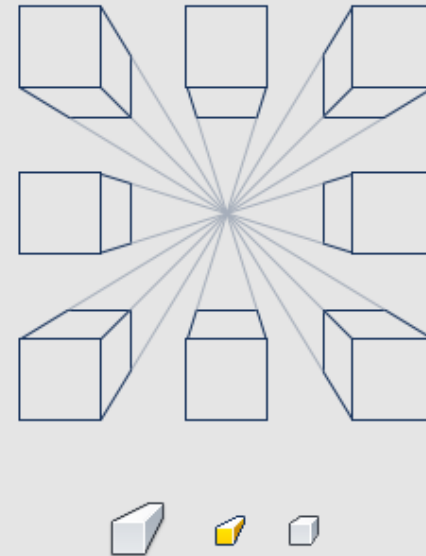
3) Horizontal oblique bimetric projection



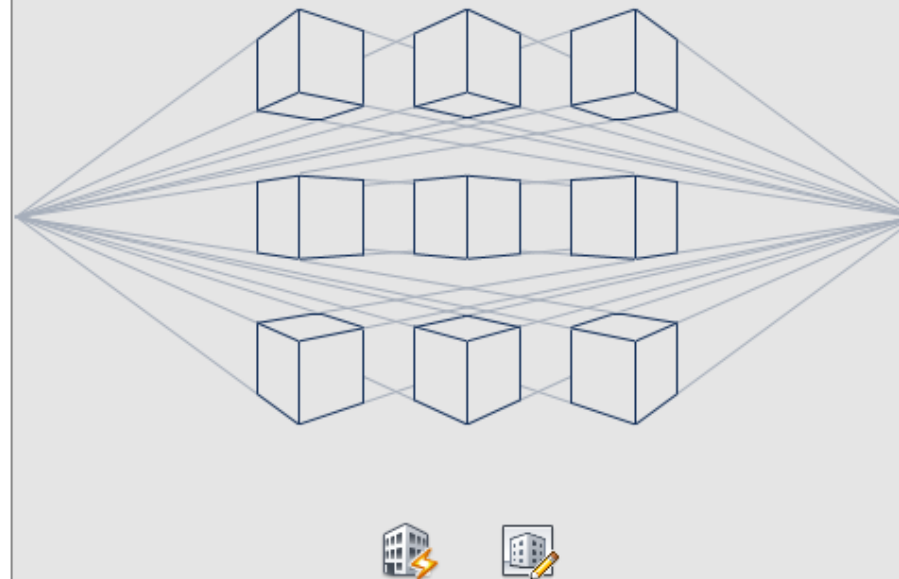
Perspective

2. Perspective

1) One-point perspective (parallel perspective)



2) Two-points perspective (angular perspective)



Perspective

3) Three-points Perspective



Style

1. Corner Smoothing

Decrease the Transparency of outer 90 degree corners by 41%, or set "Storke" as the follows figure in the "Layer Style" in Adobe Photoshop.

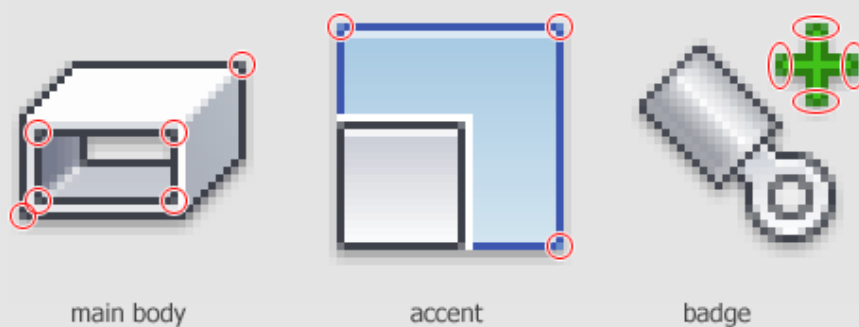
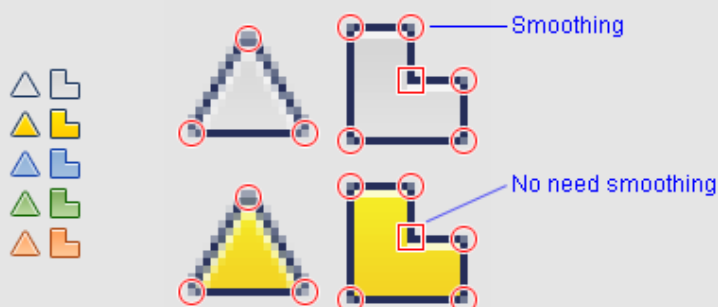
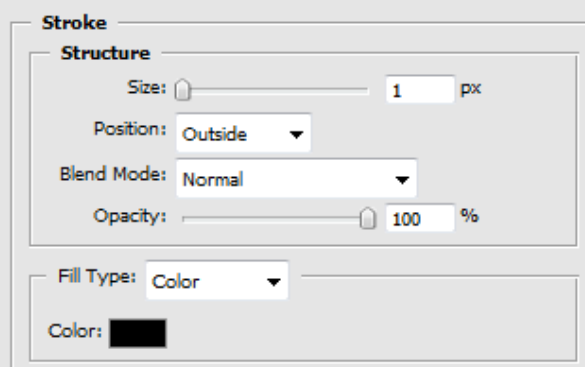
Applies to all icon sizes and all elements. (inculde main boxy, accent and bardges)

Do not apply to objects which are smaller then 6x6 pixels.



Style

Set Stroke in Layer style

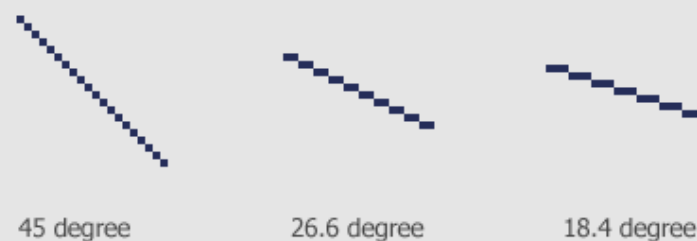


Style

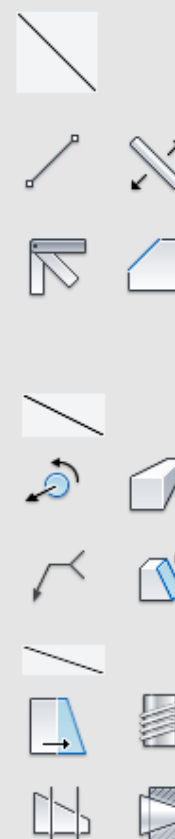
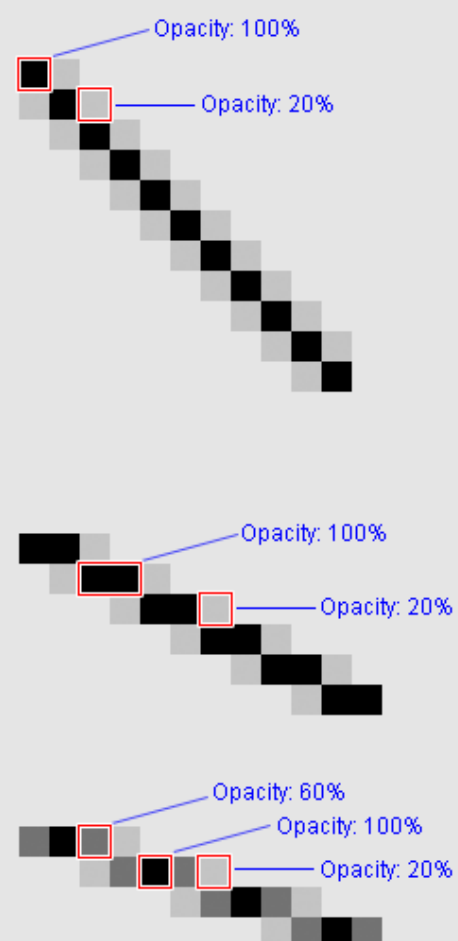
2. Anti-Aliasing & Crisp outlines

1) Straight Line

Normal draw the straight lines are horizontal, vertical, 45°, 26.6° and 18.4° (include 0 (or 90,180,270) degree $\pm$ 26.6 (or 18.4) degree). Usually, we can draw 26.6° instead of the 30° and use 18.4° instead of the 20° if the angle values are allown not precision.



Following are Anti-Aliasing 45, 26.6 and 18.4 degree lines

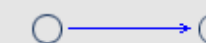
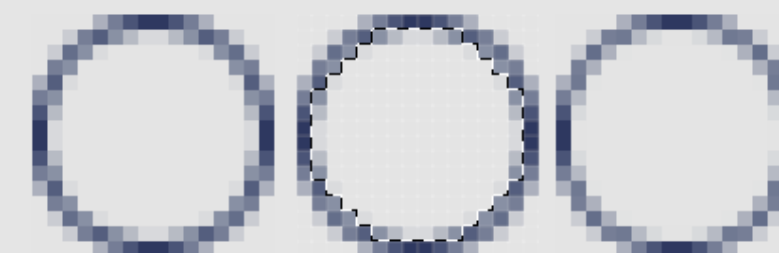


Style

2) Circle

The general anti-alias circle are not crisp enough. We can use the following steps to create a crisp circle in Adobe Photoshop.

1. Create a stroke circle you need, diameter value is "D" px
2. Create a circle marquee which diameter value is "D-2" px and the same center point with the stroke circle.
3. Delete the stroke circle in the circle marquee.

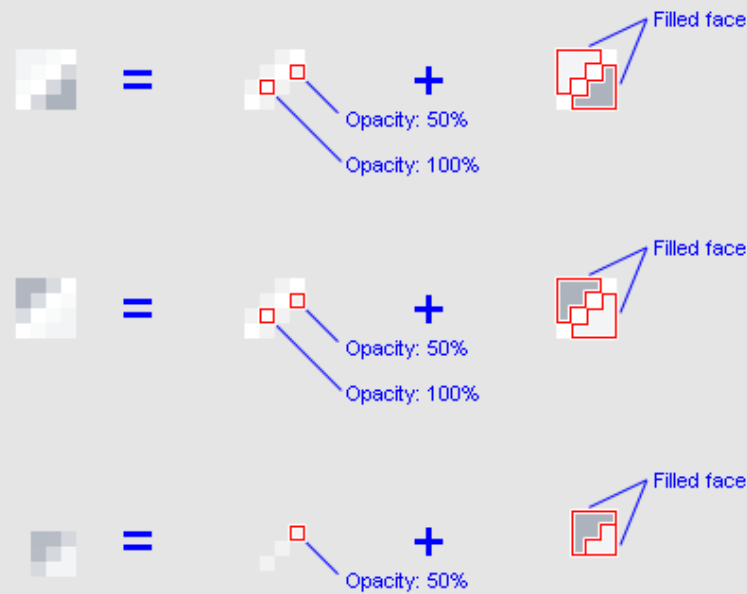


3) Face

Set the anti-alias effect between two filled faces if the icon size is larger than 24x24 pixles.

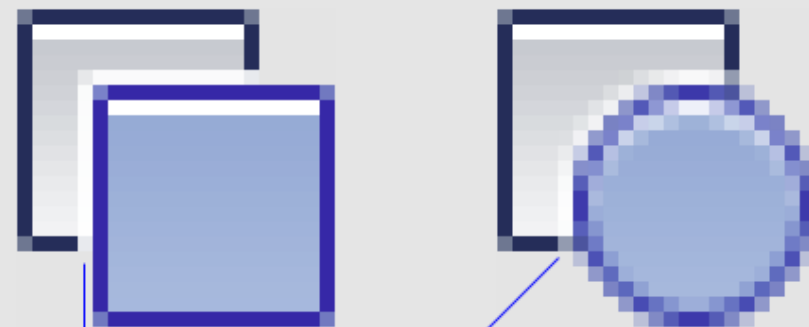


Style



3. Overlying Elements

Separate elements sitting on top of each other should be separated with a 1 pixel white and anti-aliased line with 80%-90% transparency. Usually you can set the white line by using Layer Style in the Adobe Photoshop. Apply for the mainbody, accent and badge.



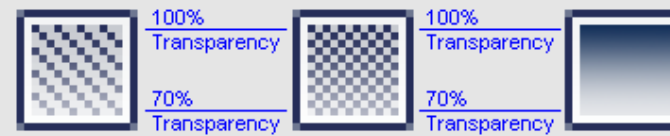
White line 80%-90% opacity
Width 1 px



Style

4. Mapping

Mappings are include Hatching, Table lines, UV curves, list iteams and etc. Mappings are faded towards their icons: Top are 100% transparency and the bottom are 70% transparency.



Loose hatching style

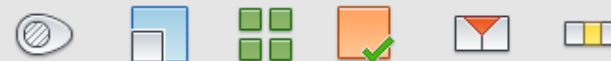
Dense hatching style

Solid hatching style



5. Outline

All of the primary outline forms within an icon and its elements are rendered with a dark 1 pixel outline. They have been defined in color section. Never use a color gradients within an outline.



6. Dashed and Center Lines

1) Dashed lines: Use a 2 px on + 1 px off dashed line. Applies to all icon sizes.



2) Center Lines: Use a 2 px on + 2 px off dashed line stand for the short line and the spaces at left and right. The short line and spaces can lagrer than 2 px if it more clearly. At least draw one short line and two spaces in an icon.

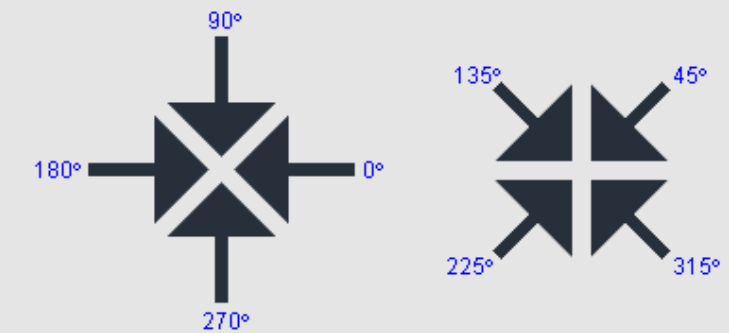


Style

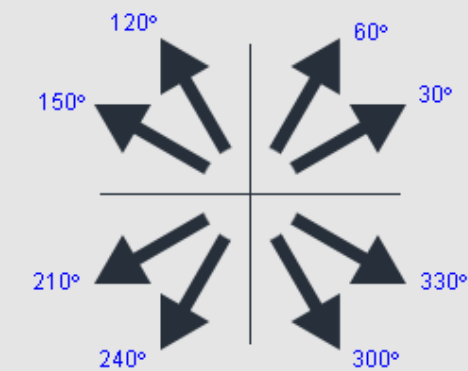
7. Arrows

1) Arrowheads:

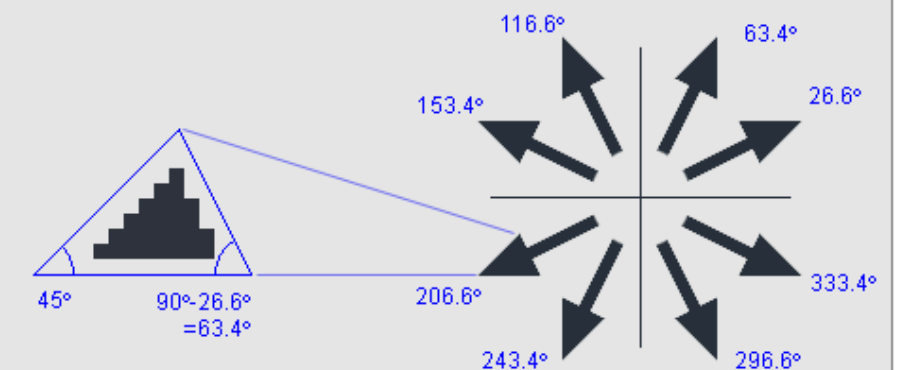
(1) The arrowheads is a right angled isosceles triangle if the angle values of the arrowlines are 0 (90,180,270) degree or 45 (135,225,315) degrees.



(2) The arrowheads is a equilateral triangle if the angle values of the arrowlines are $30 (0 \pm 30, 90 \pm 30, 180 \pm 30, 270 \pm 30)$ degrees.



(3) The arrowheads is the following shape if the angle values of the arrowlines are 26.6 (0 ± 26.6 , 90 ± 26.6 , 180 ± 26.6 , 270 ± 26.6) degrees.

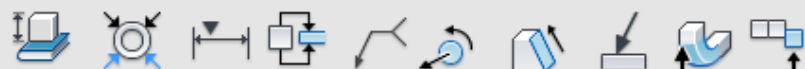




Style

2) Arrowlines: The general width of the arrowlines is 1 px. You can use more widely arrowlines if it's consistency.

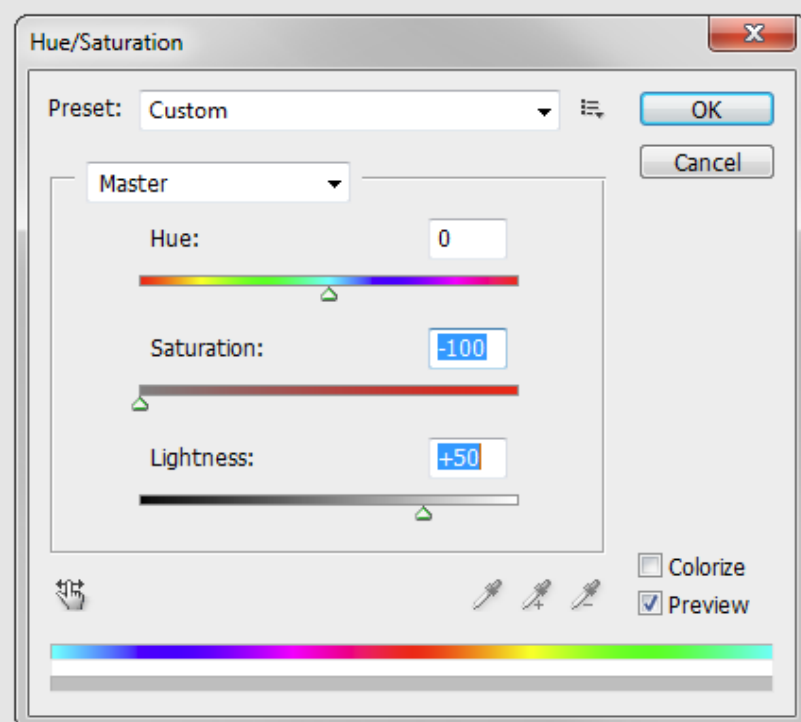
3) Anti-alias: Apply all arrows anti-alias effect if the size of the arrowhead are bigger than 5x5 px.



8. Disable States

Create a disable state icon base an existing icon in Adobe Photoshop:

1. Set Saturation -100
2. Set Lightness +50



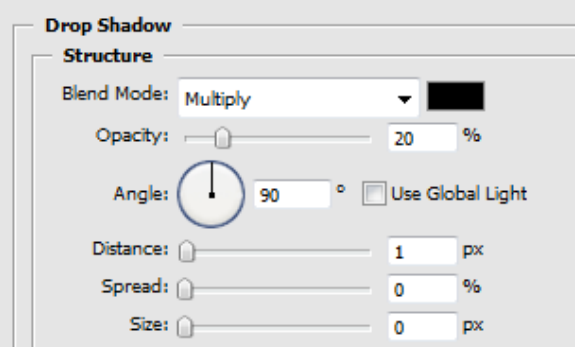
Effects

The additional effects are used to accentuate the icons which placed in the Inventor UI. The main effects are: Shadows and Protrude (white shadows).

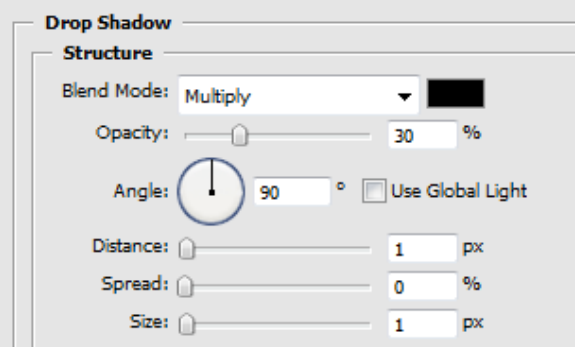
1. Shadows

This effect normal apply for Ribbon icons.

- 1) Do not apply shadows to icons equal to or smaller than 16x16 px.
- 2) 24x24 px icons use the following Adobe Photoshop setting:



- 3) 32x32 px and larger icons use the following Photoshop Setting:

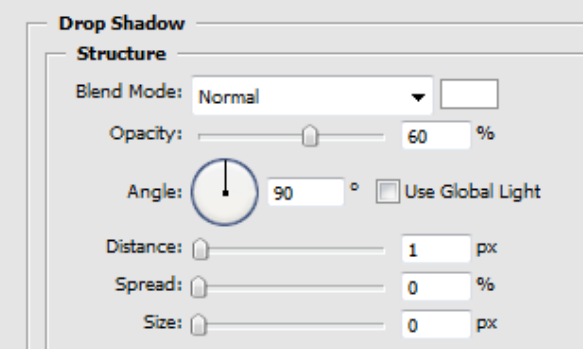


Note: The icon demensions are include shadow sizes.

Effects

2. Protrude (White shadow)

This effect naormal apply for Mini-tool bar icons. 16x16 px and 24x24 px icons use the following Adobe Photoshop setting:



Note: The icon demensions are not include the Protrude effect. It means this icon demensions are 16x17 px and 24x25 px.



Format

There are 6 type icons used in Inventor. They are icons for: Ribbon, Dialogue, Browser, Context menu, Mini-toolbar (In Canvas UI) and Marking menu.

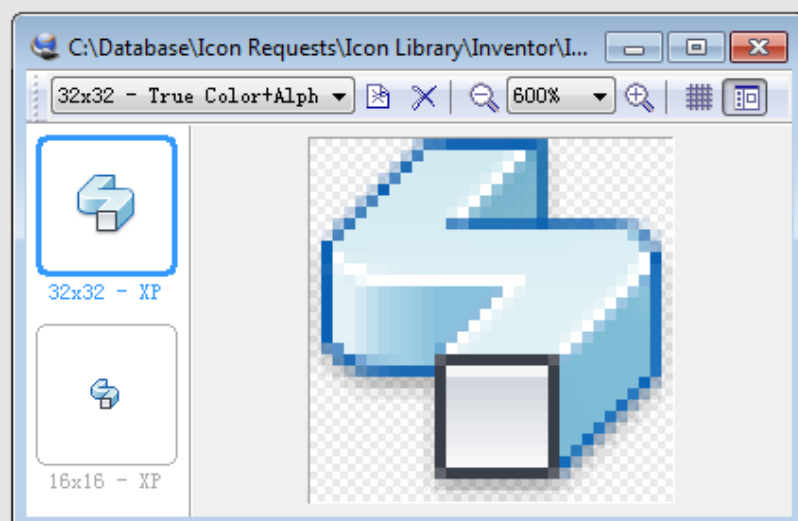
1. The format for them are ruled in the follow table:

	Ribbon	Dialog	Browser	Context menu	Mini-toolbar (In canvas UI)	Marking Menu
Canvas Size (px)	16x16 32x32	16x16	16x16	16x16	16x17 24x25	16x16 32x32
Resolution (dpi)	72	72	72	72	72	72
Format	.ico	.ico	.bmp	.ico .png	.png	.ico
Color Sheme	Amber & Cobalt	Amber	Amber	Amber	Amber & Cobalt	Amber & Cobalt
Comments	Two icons in one .ico file		Use magenta color (#ff00ff) as the transparent area			

Notes:

1) Ribbon & Marking Menu

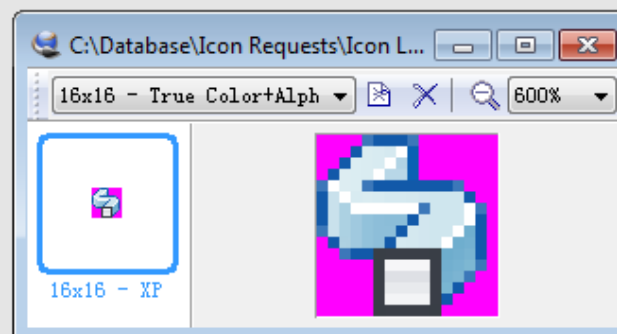
Icons for Ribbon and Marking Menu use combined ICO files. (combine 32 and 16 icons in one ICO file)



2) Browser

We use pure Magenta color (RGB: 255,0,255; #ff00ff) as the transparent area in the browser icons. It means the magenta color region will be remove from this icon when it in the application. So should never use the magenta color to draw a icon.

Format



2. Naming rule

1) Icon for Ribbon and Marking Menu.

INV_Environment_Command(FEATURE)_Cmd_Dimensions.ico

- INV_2DSketch_Polygon_Cmd_32x32.ico
- INV_Assembly_AddAssembly_Cmd_32x32.ico
- INV_DrawingViews_DetailView_Cmd_16x16.ico
- INV_SheetMetal_Bend_Cmd_16x16.ico
- INV_View_PerspectiveCamera_Cmd_16x16.ico

2) Icons for Dialog and Mini-toolbar.

INV_Environment_Command(FEATURE)_Function.ico / .png

- INV_Model_Chamfer_tangentialyedge.png
- INV_Model_Draft_fixplane.png
- INV_Model_Fillet_Setbacks.png

3) Icons for Browser.

INV_Environment_Command(FEATURE)_Function_Status_Dimensions.bmp

- INV_Model_Extrude_Active_Browser_16x16.bmp
- INV_Model_ExtrudeSurface_Active_Browser_16x16.bmp
- INV_Model_ExtrudeSurface_Active_Shared_Browser_16x16.bmp

Examples in Context

